MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – TEST (SET – B)
Chapter: Areas Related to Circles

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory.
2. The question paper consists of 11 questions divided into three Sections: A, B, and C.
3. **Section A** comprises 5 Multiple Choice Questions (MCQs) of 1 mark each.
4. **Section B** comprises 3 Short Answer questions of 2 marks each.
5. **Section C** comprises 3 Long Answer questions of 3 marks each.
6. Use of calculator is not permitted.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. What is the length of the arc of a sector of a circle with radius 14 cm and central angle 90°? [1]

(A) 22 cm (B) 44 cm (C) 88 cm (D) 11 cm

Q2. A sector of a circle with central angle 120° and area $\frac{264}{7}$ cm^2 is cut from a circle. The radius of the circle is: [1]

(A) 6 cm (B) 5 cm (C) 7 cm (D) 12 cm

Q3. If the sector of a circle with diameter 14 cm makes an angle 90° at the centre, then the perimeter of the sector is: [1]

(A) 25 cm (B) 11 cm (C) 36 cm (D) 22 cm

Q4. If the area of a sector is one-twelfth that of a complete circle, then the central angle of the sector is: [1]

(A) 36° (B) 30° (C) 60° (D) 45°

Q5. If an arc subtends an angle of 90° at the centre of a circle, then the ratio of its length to the circumference of the circle is: [1]

(A) 2 : 3 (B) 1 : 4 (C) 4 : 1 (D) 1 : 3

SECTION B: Short Answer Questions (3 × 2 = 6 Marks)

Q6. A chord of a circle of radius 14 cm subtends an angle of 90° at the centre of the circle. Find the area of the corresponding minor segment of the circle. $\left[\text{Use } \pi = \frac{22}{7}\right]$ [2]

Q7. From a circular sheet of paper of radius 70 cm, a quadrant is cut out. Find the area of the remaining sheet. $\left[\text{Use } \pi = \frac{22}{7}\right]$ [2]

Q8. Three sectors of a circle of radius 5 cm, having central angles 35°, 50°, and 95° at the centre, are shaded. Find the total area of the shaded region. $\left[\text{Use } \pi = \frac{22}{7}\right]$ [2]

SECTION C: Long Answer Questions (3 × 3 = 9 Marks)

Q9. A chord of a circle of radius 14 cm subtends an angle of 60° at the centre. Find the area of the corresponding minor segment of the circle. $\left[\text{Use } \pi = 3.14 \text{ and } \sqrt{3} = 1.73\right]$ **[3]**

Q10. An arc of a circle of radius 10 cm subtends a right angle (90°) at the centre of the circle. Find the area of the corresponding major sector. [Use $\pi = 3.14$] **[3]**

Q11. A sector is cut from a circle of radius 21 cm. The central angle of the sector is 150°. Find :

(i) the length of the arc of this sector, and

(ii) the area of the sector. $\left[\text{Use } \pi = \frac{22}{7}\right]$

[3]

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